ST443 Week 2-3

- Machine learning principles
- Intro to linear regression (Chapter 2-3)

8 October 2026

The slides may be updated up to, and after the seminars.

ST443 seminar slides



ML principals

- Basic principals
- ML workflow
- Exercise (bias-variance decomposition, risk minimisation)
- Problem set



Which of these is true about machine learning (ML)?

A. Making decisions based on data

B. ML models identify patterns and make predictions by learning from examples.

C. It is desirable that decisions, predictions, or behaviours of a ML model can be understood and interpreted by humans.



Which of these is true about machine learning (ML)?

A. Making decisions based on data
B. ML models identify patterns and make predictions by learning from examples.
C. It is desirable that decisions, predictions, or behaviours of a ML model can be understood and interpreted by humans –

This describes the field of Explainable AI (XAI). XAI is important in high-stakes applications e.g healthcare



T/F

A ML model is fitted to the test data and evaluated on the training data





T/F

The ultimate goal of ML is to:

A. minimise the training error
B. minimise the generalisation (test) error

Objective of machine learning



Key idea: the real learning objective

Our goal is to minimize **error on unseen future examples**.

Average error over all possible unseen examples from the joint distribution $P(X, Y)$.



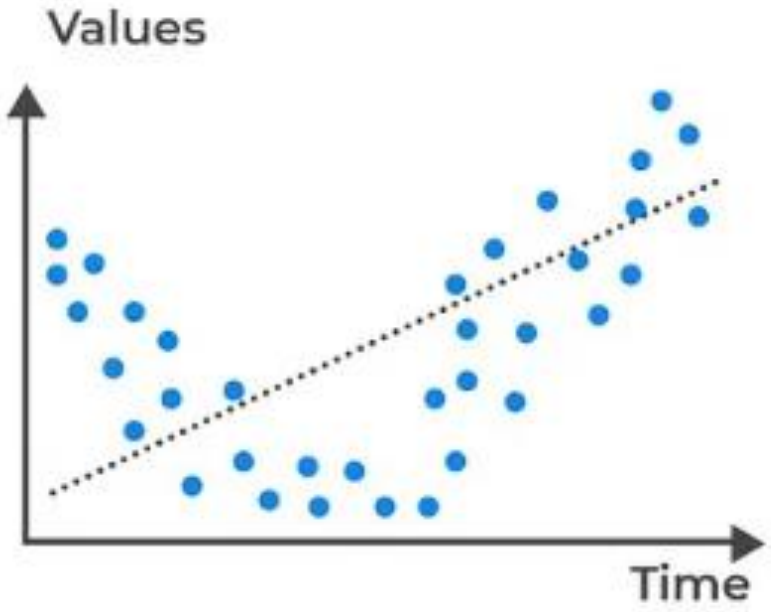
T/F

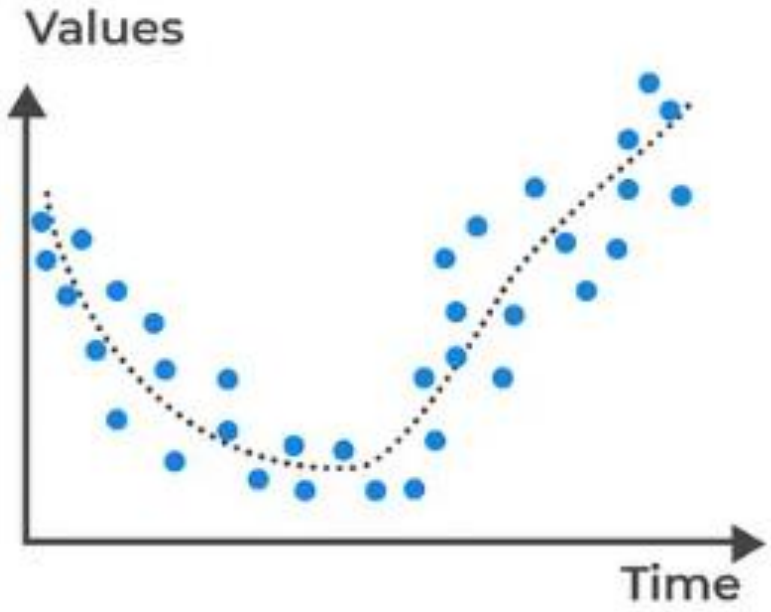
A ML that minimises the training error will minimise the generalisation (test) error.

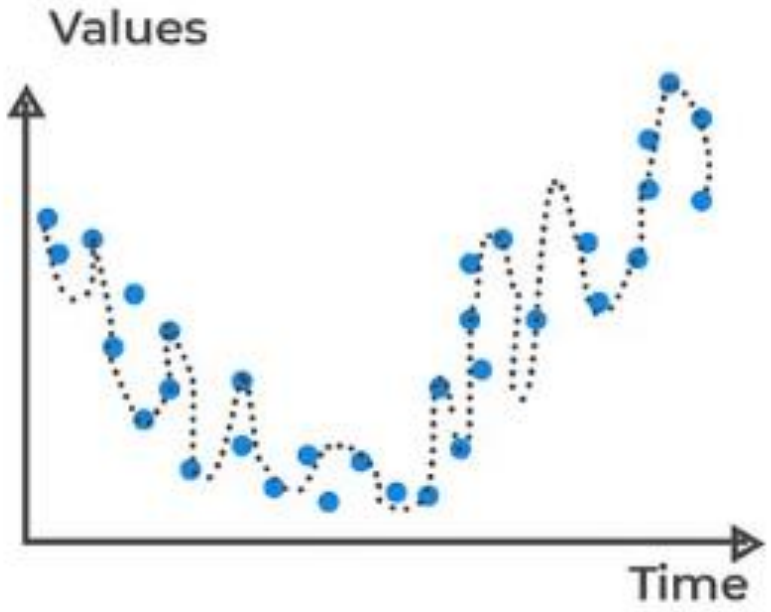


- Which plot shows the model fits best to the training data
- And which plot show the model that has the lowest generalisation error?

Plots show the training data, and the fitted model









T/F

If a model learned from training data approximates the true underlying pattern well, it will predict well on test data from the *same distribution*

Types of generalisation

Suppose the training data is sampled i.i.d. from
$(\mathbf{X}_1, y_1), (\mathbf{X}_2, y_2), \ldots, (\mathbf{X}_n, y_n) \sim F$

and the test data is sampled i.i.id from
$(\mathbf{X}'_1, y'_1), (\mathbf{X}'_2, y'_2), \ldots, (\mathbf{X}'_{n'}, y'_{n'}) \sim F'$

In-distribution (ID) generalization: $F = F'$

Called the “i.i.d assumption”. Train and test distribution match

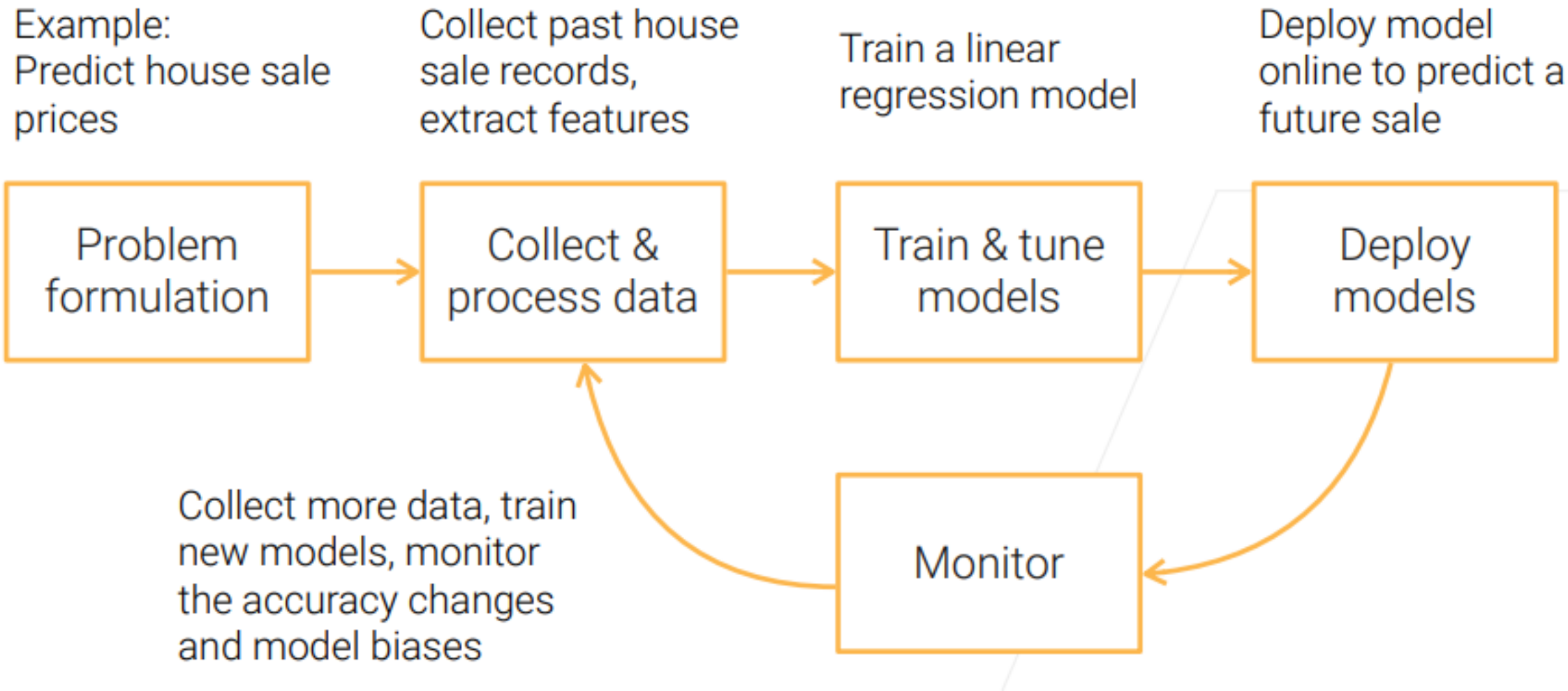
Out-of-distribution (OOD) generalization: $F \neq F'$

Train and test distribution do not match

ML principals

- Basic principals
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- Problem set

ML workflow


Example:
Predict house sale prices
Collect past house sale records, extract features
Train a linear regression model
Deploy model online to predict a future sale
Problem formulation
Collect & process data
Train & tune models
Deploy models
Monitor
Collect more data, train new models, monitor the accuracy changes and model biases

ML design

The class of functions from which the ML procedure must pick one to fit the data

What it means to *fit the data*: a function that is high for bad fits, low for good fits

How to search for the function that best fits the data



ML Design (hypothesis class, loss function, optimizer, hyperparameters, features, …)

https://www.seas.upenn.edu/~cis5190/fall2025/docs_f25/L01.pdf

Example: Simple linear regression

Hypothesis class
Linear regression

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

optimiser

We derive estimators of β_0 and β_1 using least squares estimation (introduced in Chapter 7). The least squares estimators (LSEs) of β_0 and β_1 are the values of (β_0, β_1) at which the function:

$$L(\beta_0, \beta_1) = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

Loss function

obtains its minimum.

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i (y_i - \bar{y})}{\sum_{i=1}^{n} x_i (x_i - \bar{x})} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \quad \text{and} \quad \widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}.$$

ST102 Lectures on Moodle



- What is the **risk function** in machine learning optimisation?
- For what purpose does it serve?

Risk function = the expected loss of a specific model.

It represents the long-term average loss over the probability distribution of the data, giving a sense of how well a model is expected to perform across different data points.

- For a given loss function, ℓ, the risk of a learning function f is defined by the expected loss

$$R(f) = \mathbb{E}_{X,Y}[\ell(Y, f(X))].$$

$(X, Y) \sim$ unseen data from the population

Bayes and risk

Concept of risk is central to the Bayes decision theory framework

Goal of machine learning

- For a given loss function, ℓ, the risk of a learning function f is defined by the expected loss

$$R(f) = \mathbb{E}_{X,Y}[\ell(Y, f(X))].$$

Goal of ML: find the model f that minimises the expected loss R(f)

Issues:

- The model class to search for f
- We don't know true joint distribution X,Y in the expectation

Goal of ML: find the model f that minimises the expected loss R(f)

- Issue 1: The model class to search for f
  - Solution: restrict the class (e.g. linear models, tree models,…)

Goal of ML: find the model f that minimises the expected loss R(f)

- Issue 2: In practice we cannot compute risk as we do not know the true joint distribution of X, Y.

- We can estimate risk for a given f, or provide theoretical bounds

- The training error provides an optimistically biased estimate of the expected loss due to the optimisation procedure.
  - To get an unbiased estimate of the expected loss, evaluate the model on data not used in training $(\tilde{x}_1, \tilde{y}_1), \ldots, (\tilde{x}_m, \tilde{y}_m)$. We call this the test set.
  - The test error for fitted model $\hat{f}_\theta$ is:

$$\text{Test error} = \frac{1}{m}\sum_{i=1}^{m} \ell(\tilde{y}_i, \hat{f}_\theta(\tilde{x}_i))$$

  - The test error provides an unbiased estimate of the expected loss when the test set is drawn from the same distribution as the training set.



Bayes risk

T/F

A. Bayes risk is the minimum risk achievable by any model for a given loss function

B. Bayes risk is a concept that applies to the regression problem, and the classification problem

C. Excess risk is defined as the risk of predictor f, and Bayes risk

Bayes risk

A. Bayes risk is the minimum risk achievable by any model for a given loss function

B. Bayes risk is a concept that applies to the regression problem, and the classification problem

C. Excess risk is defined as the difference between risk of predictor f, and Bayes risk:

Excess risk = $R(f) - R(f_{optimal_model})$

Supervised learning|
Regression and classification tasks

CLASSICAL MACHINE LEARNING

Data is pre-categorized or numerical

Predict a category

CLASSIFICATION

«Divide the socks by color»



Predict a number

REGRESSION

«Divide the ties by length»



Data is not labeled in any way

Divide by similarity

CLUSTERING

«Split up similar clothing into stacks»



Find hidden dependencies

ASSOCIATION

«Find what clothes I often wear together»



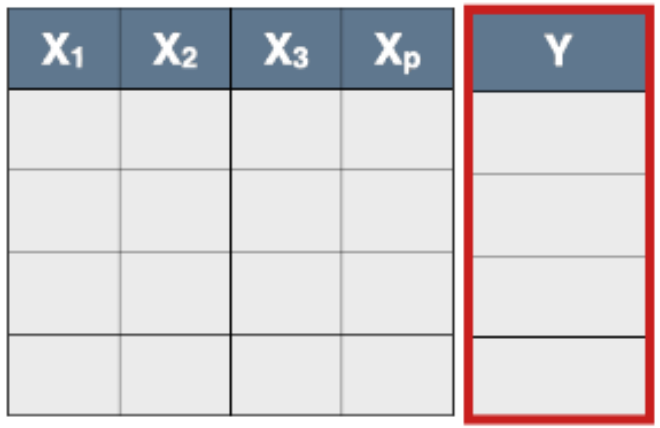
Identify sequences

DIMENSION REDUCTION (generalization)

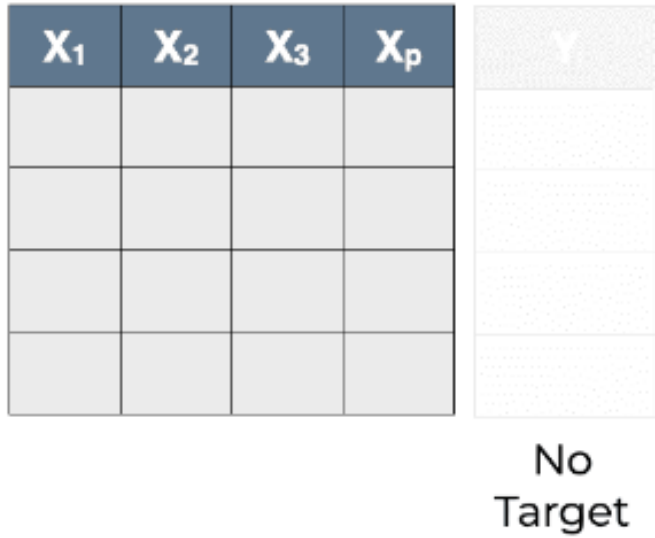
«Make the best outfits from the given clothes»



Supervised Learning



Un-Supervised Learning



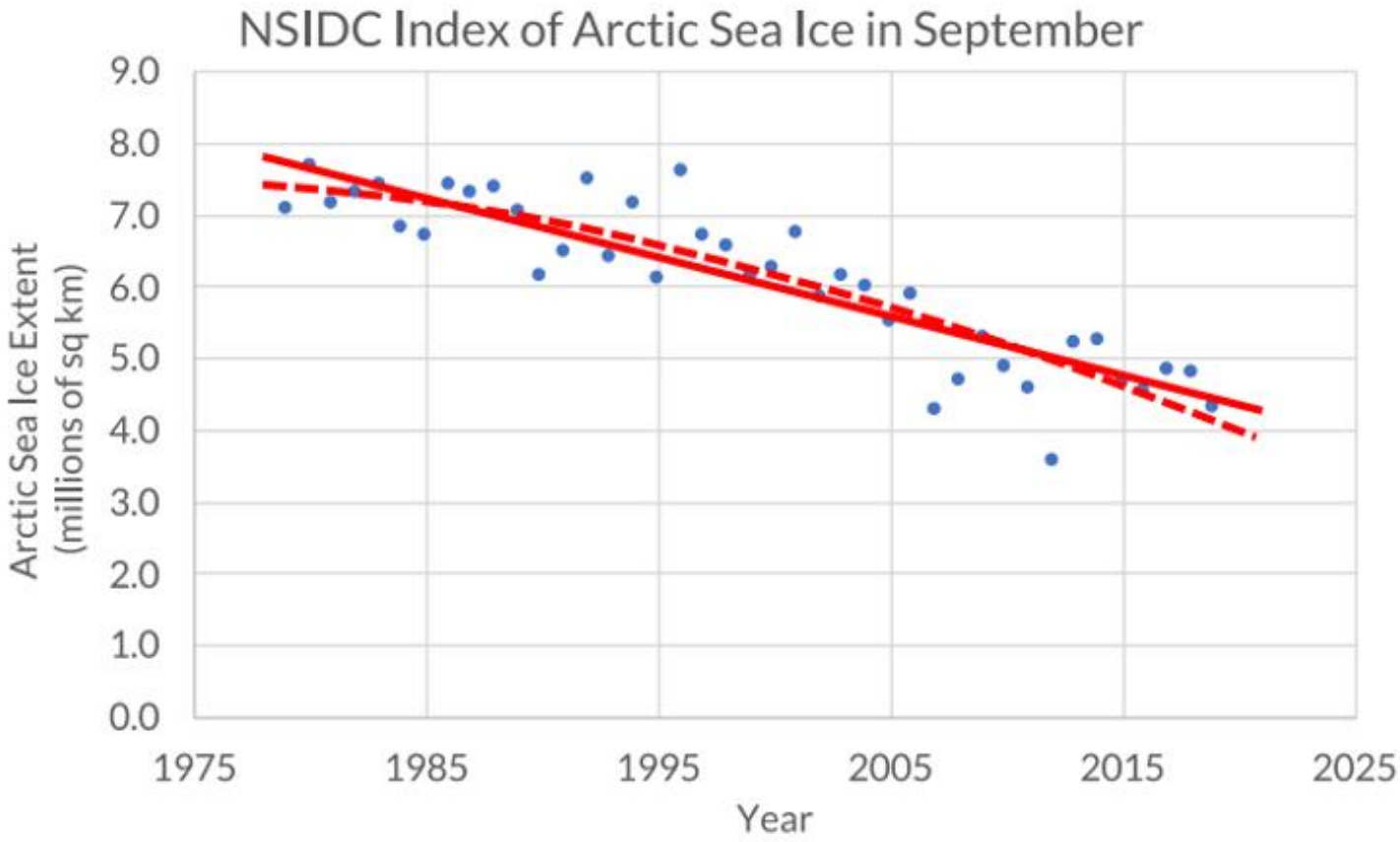


Regression and classification come under supervised learning.
How can you tell whether a task is either regression or classification?

Supervised learning: regression task

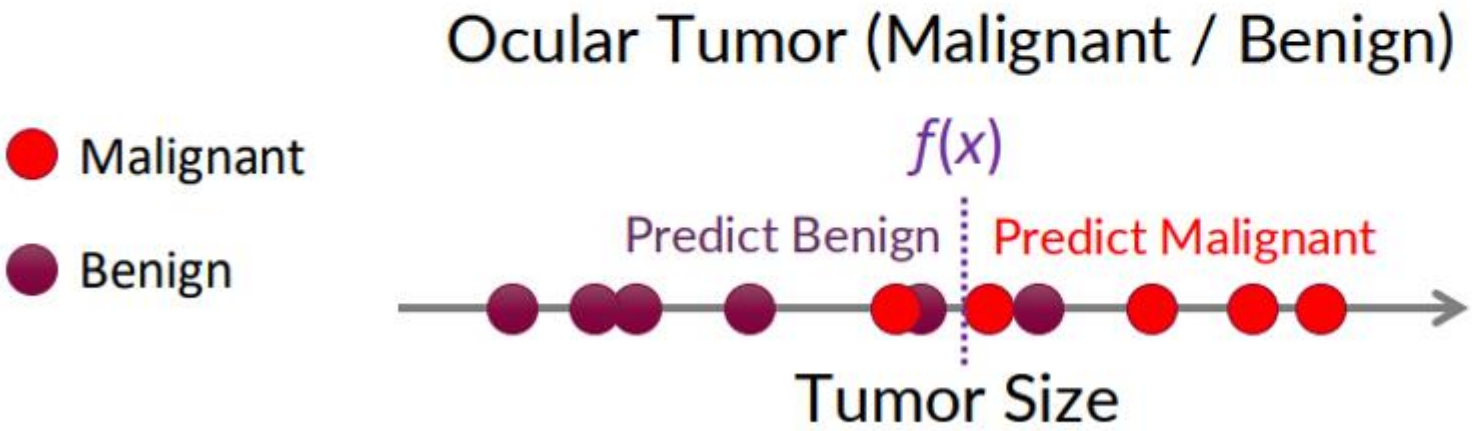
- Given $D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$
- Learn a function $f(x)$ to predict y given x
  - y is numeric == regression





Supervised learning: classification task

- Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
  - y is categorical == classification



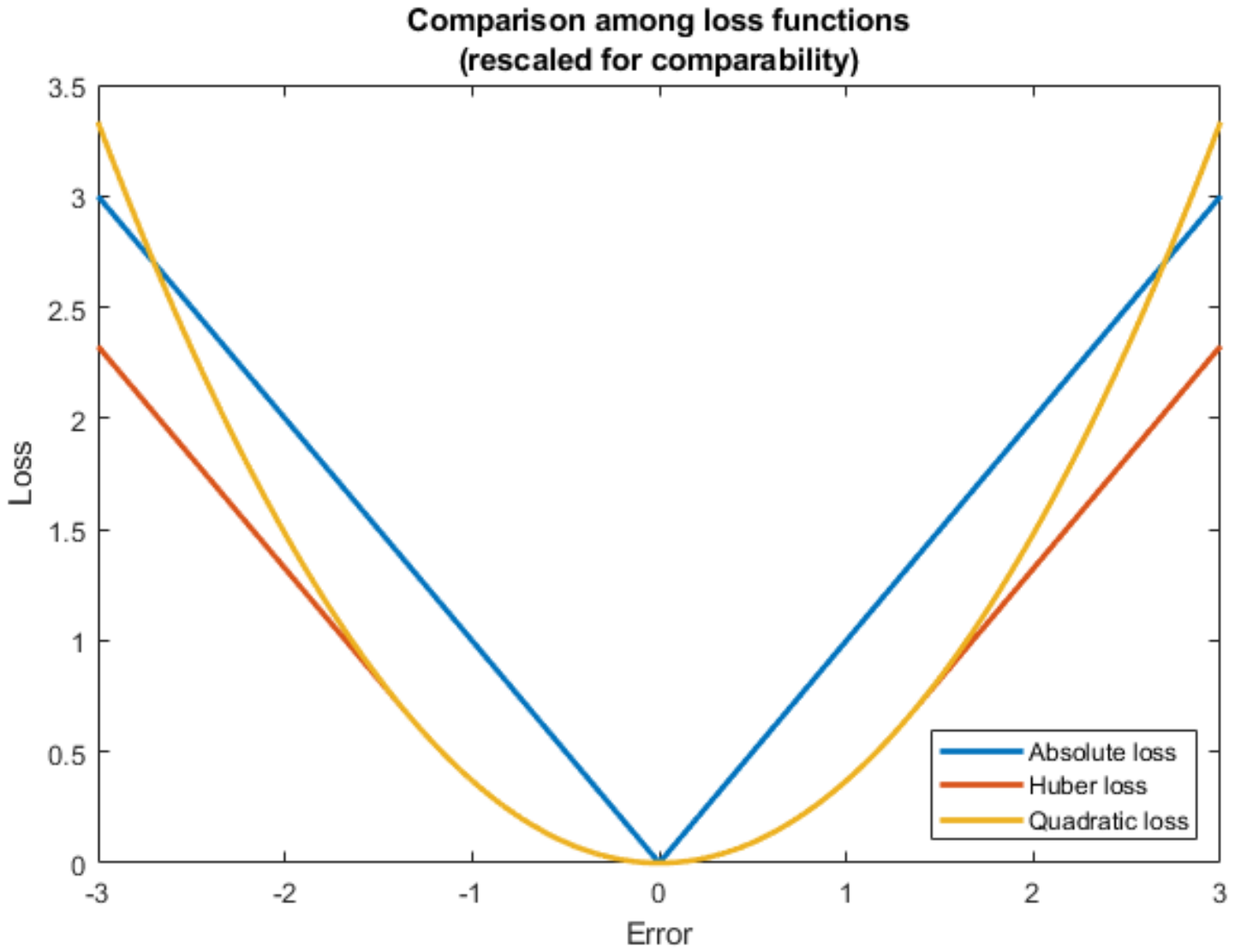




Name some loss functions commonly used for regression tasks

Loss functions for regression

- Squared error loss
- Absolute error loss





Regression

	Loss function	**Bayes (optimal) predictor** Achieves Bayes risk (minimiser of risk)
(a)	Squared error loss	
(b)	Absolute error loss	

Regression

	Loss function	**Bayes (optimal) predictor** Achieves Bayes risk (minimiser of risk)
(a)	Squared error loss	$f(x) = E(Y\|X = x)$
(b)	Absolute error loss	$f(x) = \text{ median of } p_{Y\|X=x}(y)$



Name some loss functions commonly used for binary classification tasks

Loss functions for binary classification

True class y (0 or 1), predicted class $\hat{y}$ (in [0,1])

- Zero-one loss:

$$\ell(y, \hat{y}) = 1_{\left\{\left(y-\frac{1}{2}\right)\left(\hat{y}-\frac{1}{2}\right)<0\right\}}$$

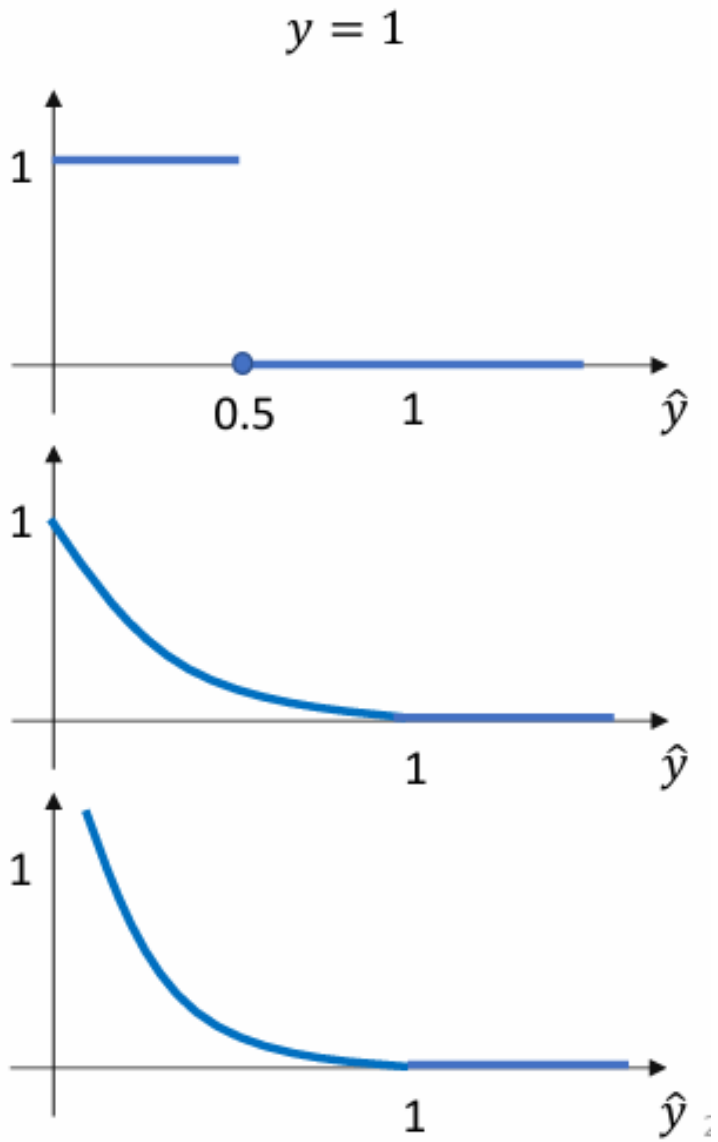
- Squared loss:

$$\ell(y, \hat{y}) = (\hat{y} - y)^2$$

Binary

- Cross-entropy loss: = negative log-likelihood

$$\ell(y, \hat{y}) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$





Classification

	Loss function	**Bayes (optimal) classifier** Minimiser of risk
(a)	0-1 loss	
(b)	Binary cross-entropy loss	

Classification

	Loss function	**Bayes (optimal) classifier** Minimiser of risk
(a)	0-1 loss	$f(x) \in \arg\max_y \mathbb{P}[Y = y \mid x]$
(b)	Binary cross-entropy loss	$f(x) = \mathbb{P}[Y = 1 \mid X = x] = \mathbb{E}[Y \mid X = x]$



Name some loss functions commonly used for multi-class classification problems

Name some loss functions for multi-class classification problems

- 0-1 loss
- Cross-entropy loss (= negative log-likelihood)
- KL divergence

ML principals

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Sources of prediction error

Suppose you have fitted a model for a regression task and use it to make a prediction for an input that was not in the training data.

What are the three theoretical sources of prediction error when computing the test MSE i.e MSE on unseen data point?

What are the three theoretical sources of prediction error in a regression task?

- Bias
- Variance
- Noise (irreducible error)

This holds for regression and conceptually applies to classification tasks

Bias–variance decomposition board

$y_i = f(x_i) + \varepsilon_i$, inputs fixed, noise independent, mean 0, variance σ^2; new $Y_0 = f(x_0) + \varepsilon_0$.



T/F

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)]$$

Bias–variance decomposition board

$y_i = f(x_i) + \varepsilon_i$, inputs fixed, noise independent, mean 0, variance σ^2; new $Y_0 = f(x_0) + \varepsilon_0$.

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

$$\mathbb{E}_{D,\epsilon}\left[y_0 - \hat{f}(x_0; D)\right]^2 = \overbrace{\text{Bias}_D^2[\hat{f}(x_0; D)]}^{\geq 0} + \overbrace{\text{Var}_D[\hat{f}(x_0; D)]}^{\geq 0} + \text{Var}(\epsilon),$$

$$\text{Bias}_D[\hat{f}(x_0; D)] = \mathbb{E}_D[\hat{f}(x_0; D) - f(x_0)],$$

$$\text{Var}_D[\hat{f}(x_0; D)] = \mathbb{E}_D\left(\hat{f}(x_0; D) - \mathbb{E}_D[\hat{f}(x_0; D)]\right)^2.$$



$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

In what way is the bias-variance decomposition useful?

Bias–variance decomposition board

$y_i = f(x_i) + \varepsilon_i$, inputs fixed, noise independent, mean 0, variance σ^2; new $Y_0 = f(x_0) + \varepsilon_0$.

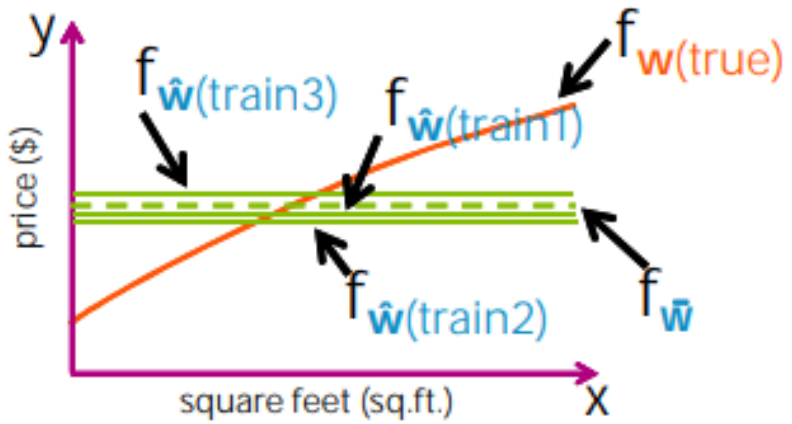
$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \mathrm{Bias}(\hat{f}(x))^2 + \mathrm{Var}[\hat{f}(x)] + \sigma^2$$

The bias-variance decomposition helps us understand how model flexibility impacts predictive performance on unseen data in a regression task.

Case: **underfitted** model (green line)

Bias contribution

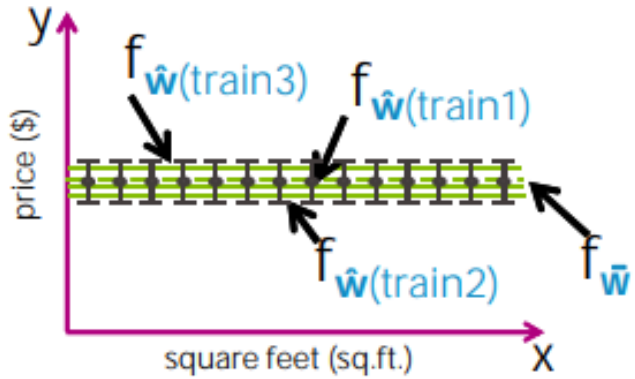
Over all possible size N training sets, what do I expect my fit to be?



low complexity → high bias

Variance contribution

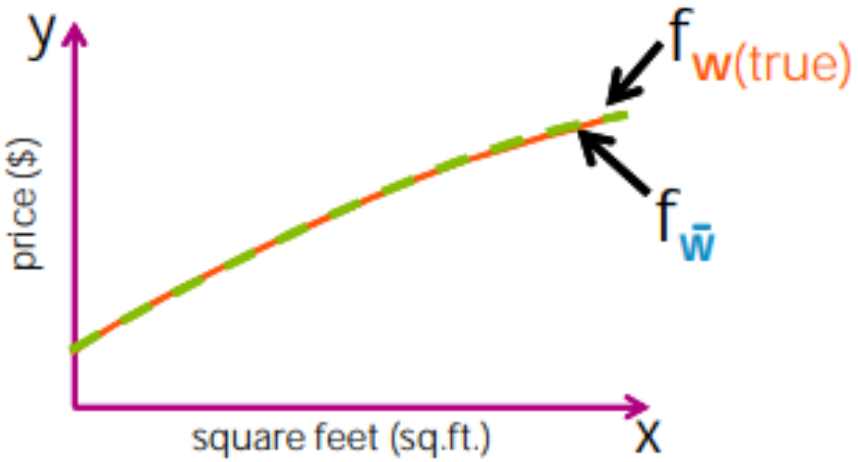
How much do specific fits vary from the expected fit?



low complexity → low variance

Case: **overfitted** model

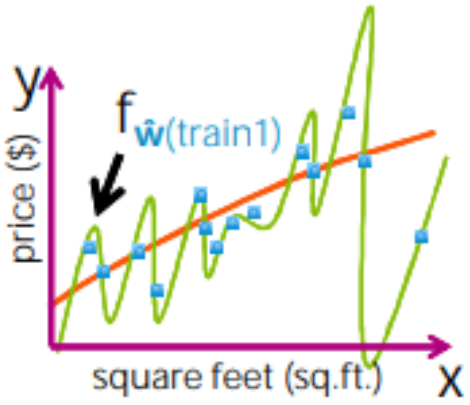
Bias of high-complexity models



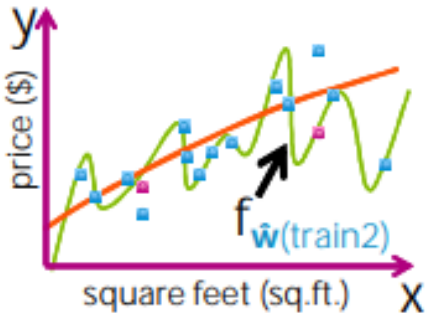
high complexity
→
low bias

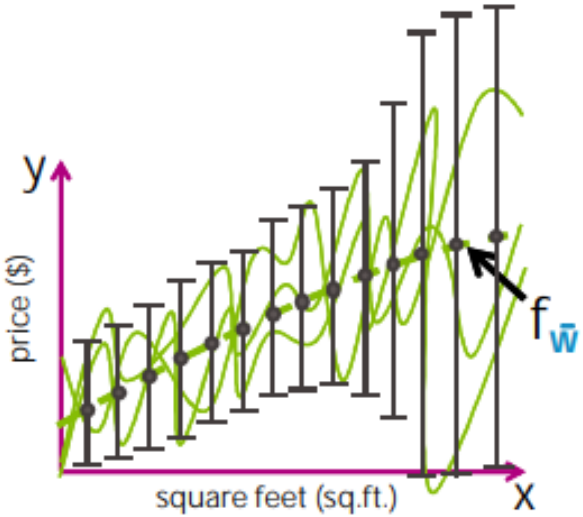
Variance of high-complexity models

Sample 1



Sample 2



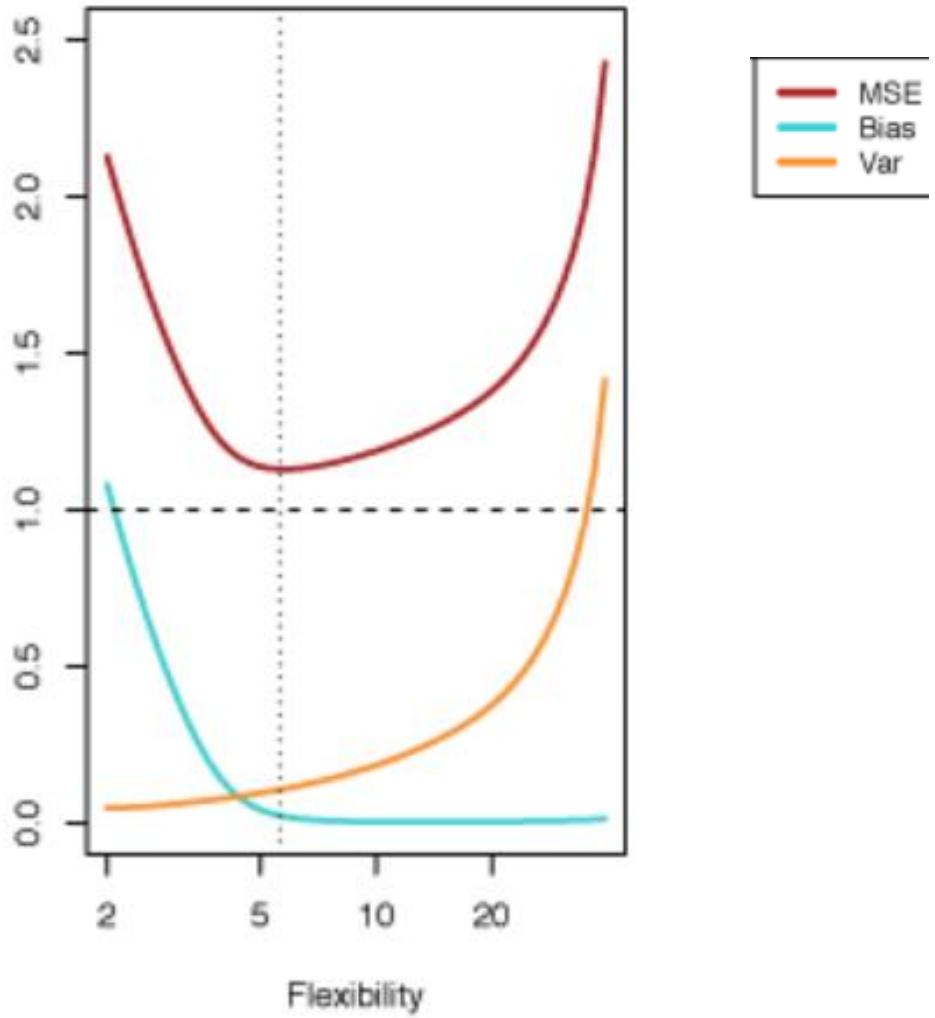


high complexity
→
high variance

Trade-off between bias and variance in expected test error

- As model complexity increases, bias of predictions on *new* data from trained model decreases, while its variance increases.

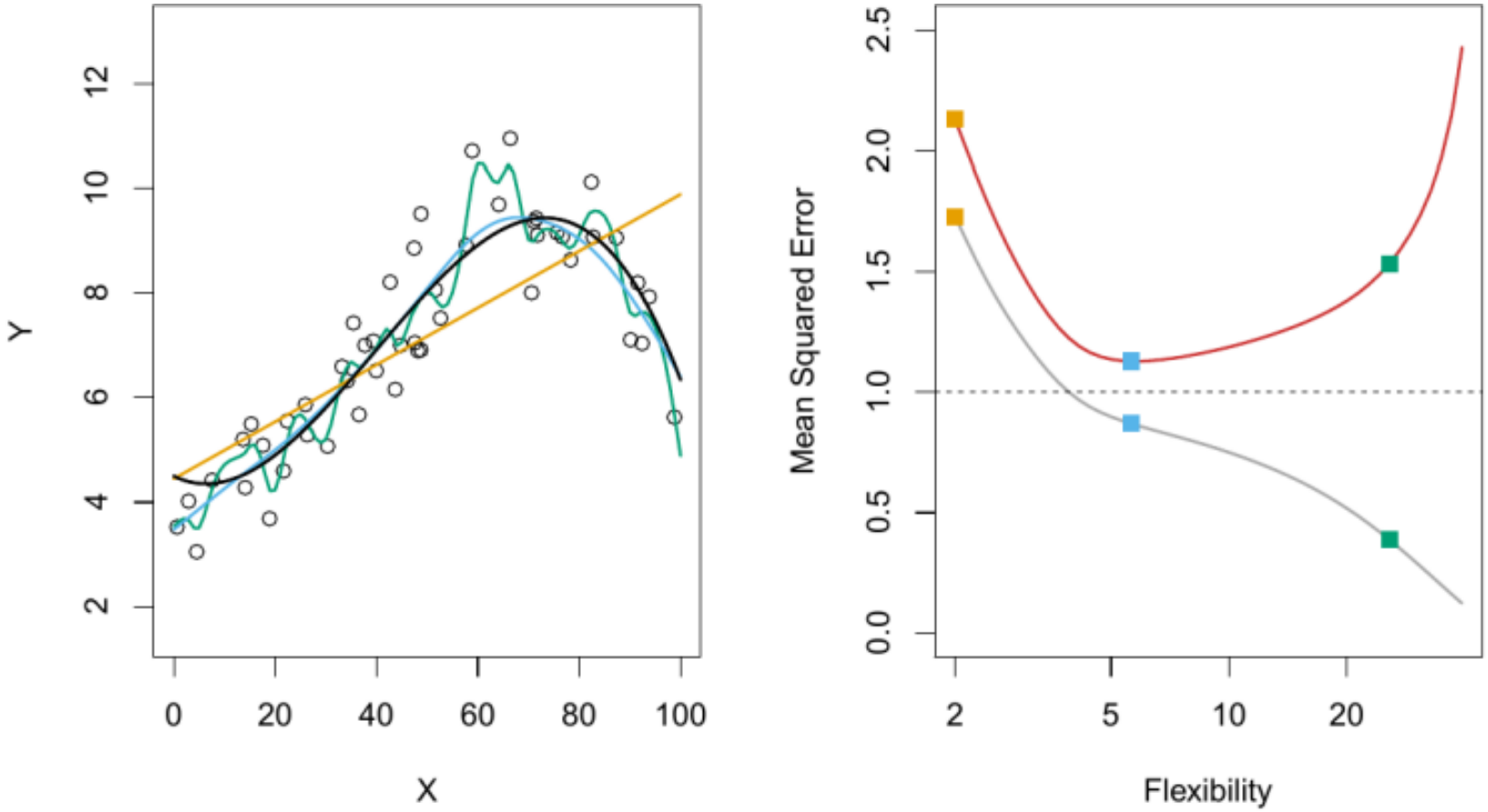
- By understanding the trade-off between bias and variance, we can manipulate model complexity to find a model that well predict well on unseen observations.





Does the bias-variance decomposition apply to training MSE?

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$



models of different flexibility [r.h.s.: training/test MSE vs flexibility]

Example. Bias-variance decomposition for k-NN regression.

For a new input x_0: find the k training inputs closest to it, $(x_{(1)}, y_{(1)}), \ldots, (x_{(k)}, y_{(k)})$, and average their outcomes

$$\hat{f}_k(x_0) = \frac{1}{k} \sum_{l=1}^{k} y_{(l)}$$

Example. Bias-variance decomposition for k-NN regression.

x	**1**	**2**	**4**	**7**	**8**
y	3	5	4	10	9

a. Compute the k-nearest-neighbor prediction at $x_0 = 3.5$ for $k = 1$, $k = 3$ and $k = 5$.

$$\hat{f}_k(x_0) = \frac{1}{k} \sum_{l=1}^{k} y_{(l)}$$

k	Nearest neighbour(s)	$\hat{f}_k$
1		
3		
5		

x	**1**	**2**	**4**	**7**	**8**
y	3	5	4	10	9

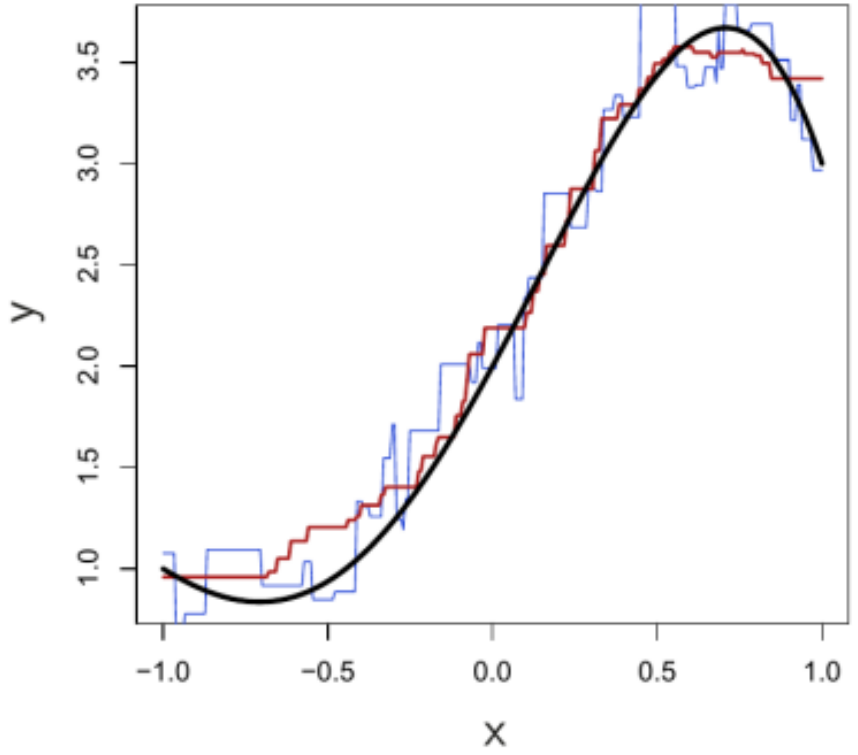
a. Compute the k-nearest-neighbor prediction at $x_0 = 3.5$ for $k = 1$, $k = 3$ and $k = 5$.

$$\hat{f}_k(x_0) = \frac{1}{k} \sum_{l=1}^{k} y_{(l)}$$

k	Nearest neighbour(s)	$\hat{f}_k$
1	x=4	4
3	x=4,2,1	4
5	x=all points	6.2

Example. Bias-variance decomposition for k-NN regression.

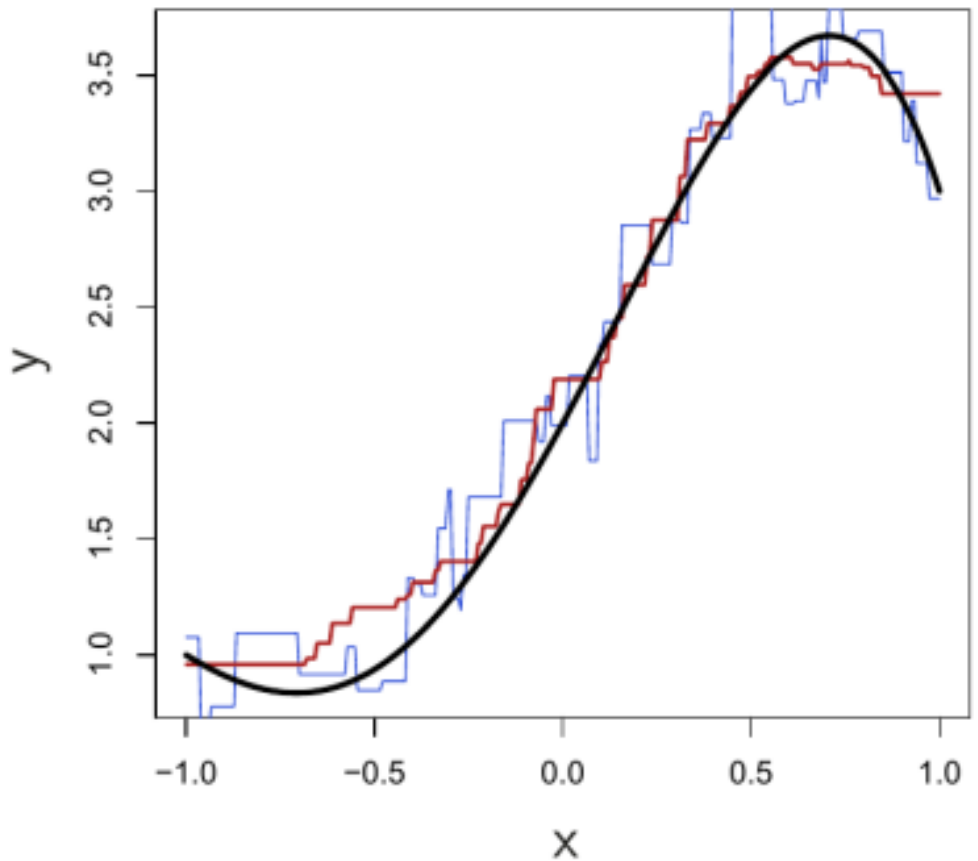
$$\mathbb{E}\left[(Y_0 - \hat{f}_k(x_0))^2\right] = \underbrace{\sigma^2}_{\text{noise}} + \underbrace{\left[f^\star(x_0) - \frac{1}{k}\sum_{l=1}^{k} f^\star(x_{(l)})\right]^2}_{\text{bias}^2} + \underbrace{\frac{\sigma^2}{k}}_{\text{variance}}$$





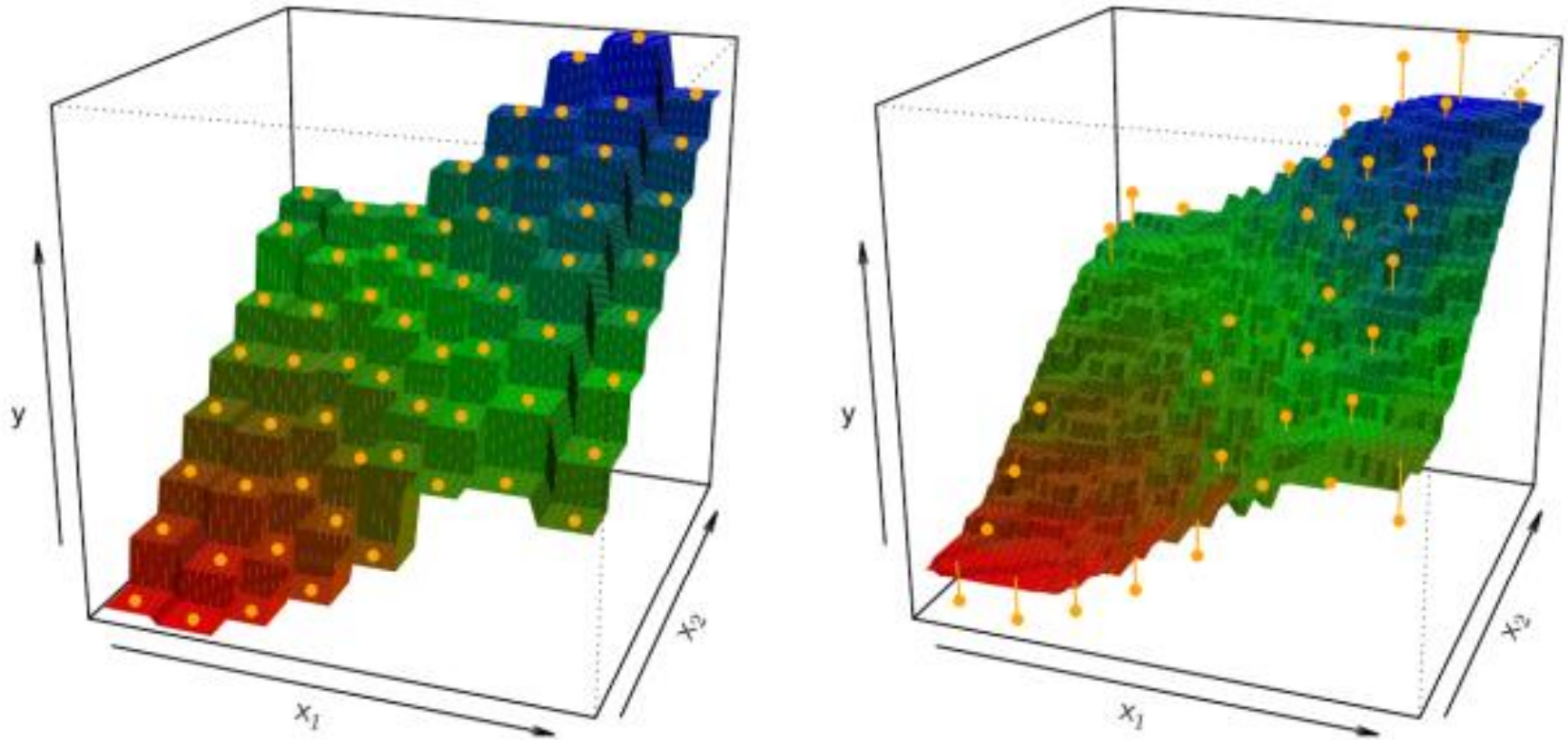
Which figure corresponds to the higher k, red or blue model?

$$\mathbb{E}\left[(Y_0 - \hat{f}_k(x_0))^2\right] = \underbrace{\sigma^2}_{\text{noise}} + \underbrace{\left[f^\star(x_0) - \frac{1}{k}\sum_{l=1}^{k} f^\star(x_{(l)})\right]^2}_{\text{bias}^2} + \underbrace{\frac{\sigma^2}{k}}_{\text{variance}}$$



fit with $K = 1$ (blue) and $K = 9$ (red);

Left: $K = 1$ (rougher), right: $K = 9$ (smoother).



Expected test loss: linear regression with squared error loss

2. **Bias-variance decomposition** Assume that $Y = f(X) + \epsilon$, where ϵ is a random variable with zero mean and variance σ^2, independent of X. Let ℓ be the squared error loss function, i.e., $\ell(y, t) = (y - t)^2$. Show that the expected test loss, conditional on $X = x$ for some arbitrary input x, can decomposed as follows:

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

where $\hat{f}$ is estimated from the training data and $\text{Bias}(\hat{f}(x)) = \mathbb{E}[\hat{f}(x)] - f(x)$.

- Additive model: $Y = f(X) + \epsilon$.
- Dataset: $D = \{(x_i, y_i)\}_{i=1}^n$.
- Expected error on an unseen x_0 [derivation for the curious-minded]:

$$\mathbb{E}_{D,\epsilon}\left[y_0 - \hat{f}(x_0; D)\right]^2 = \overbrace{\text{Bias}_D^2[\hat{f}(x_0; D)]}^{\geq 0} + \overbrace{\text{Var}_D[\hat{f}(x_0; D)]}^{\geq 0} + \text{Var}(\epsilon),$$

$$\text{Bias}_D[\hat{f}(x_0; D)] = \mathbb{E}_D[\hat{f}(x_0; D) - f(x_0)],$$

$$\text{Var}_D[\hat{f}(x_0; D)] = \mathbb{E}_D\left(\hat{f}(x_0; D) - \mathbb{E}_D[\hat{f}(x_0; D)]\right)^2.$$



What is true of the following equality

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

$$\mathbb{E}_{D,\epsilon}\left[y_0 - \hat{f}(x_0; D)\right]^2 = \overbrace{\text{Bias}_D^2[\hat{f}(x_0; D)]}^{\geq 0} + \overbrace{\text{Var}_D[\hat{f}(x_0; D)]}^{\geq 0} + \text{Var}(\epsilon),$$

A. The point $X = x$ is an arbitrary point from either the training set, or test set.

B. $\hat{f}(X)$ is estimated from the training set and it is a random variable.

C. σ^2 is the variability in Y that cannot be predicted from X (training or test point) even if we knew the true function f

What is true of the following equality

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

A. The point $X = x$ is an arbitrary point from either the training set, or test set.
   - False. It's a point from the test set

B. $\hat{f}(X)$ is estimated from the training set and it is a random variable.
   - True. While we condition on X , $\hat{f}$ depends on the training data.

C. σ^2 is the variability in Y that cannot be predicted from X (training or test point) even if we knew the true function f
   - True. The term σ^2 is called the irreducible error, noise



Notation

Classify the following as random variables vs constants.

A. $f(X)$ where X is a random variable.
B. Same as A but set X = x
C. $\hat{f}(X)$ where $\hat{f}$ is learned from the training set, and X is a point from the test set
D. Same as C but set X = x.

Classify the following as random variables vs constants.

A. $f(X)$ where X is a random variable. Random variable
B. Same as A but set X = x Constant
C. $\hat{f}(X)$ where $\hat{f}$ is learned from the training set, and X is a point from the test set Random variable (2 sources of randomness – training set, and test point)
D. Same as C but set X = x. Random variable as $\hat{f}$ depends on the training set



Bias-variance decomposition

T/F

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \underbrace{\text{Bias}(\hat{f}(x))^2}_{\geq 0} + \underbrace{\text{Var}[\hat{f}(x)]}_{\geq 0} + \sigma^2$$

$$\mathbb{E}_{D,\epsilon}\left[y_0 - \hat{f}(x_0; D)\right]^2 = \underbrace{\text{Bias}_D^2[\hat{f}(x_0; D)]}_{\geq 0} + \underbrace{\text{Var}_D[\hat{f}(x_0; D)]}_{\geq 0} + \text{Var}(\epsilon),$$

$$\text{Bias}_D[\hat{f}(x_0; D)] = \mathbb{E}_D[\hat{f}(x_0; D) - f(x_0)],$$

$$\text{Var}_D[\hat{f}(x_0; D)] = \mathbb{E}_D\left(\hat{f}(x_0; D) - \mathbb{E}_D[\hat{f}(x_0; D)]\right)^2.$$

A. Gives the generalisation error of a specific model

B. Gives the generalisation error of the learning algorithm

Bias-variance decomposition

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

$$\mathbb{E}_{D,\epsilon}\left[y_0 - \hat{f}(x_0; D)\right]^2 = \overbrace{\text{Bias}_D^2[\hat{f}(x_0; D)]}^{\geq 0} + \overbrace{\text{Var}_D[\hat{f}(x_0; D)]}^{\geq 0} + \text{Var}(\epsilon),$$

$$\text{Bias}_D[\hat{f}(x_0; D)] = \mathbb{E}_D[\hat{f}(x_0; D) - f(x_0)],$$

$$\text{Var}_D[\hat{f}(x_0; D)] = \mathbb{E}_D\left(\hat{f}(x_0; D) - \mathbb{E}_D[\hat{f}(x_0; D)]\right)^2.$$

- Dataset: $D = \{(x_i, y_i)\}_{i=1}^n$.

A. Gives the generalisation error of a specific model

- No. The expectation is over the training set so $\hat{f}$ isn't constant.

B. Gives the generalisation error of the **learning algorithm** =
on average how well do models trained by this algorithm perform
Yes.



Bias-variance decomposition

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

What is the expectation taken over?

Think about how you would use simulation to approximate the expectation on the lhs

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

$$\mathbb{E}_{D,\epsilon}\left[y_0 - \hat{f}(x_0; D)\right]^2 = \overbrace{\text{Bias}_D^2[\hat{f}(x_0; D)]} + \overbrace{\text{Var}_D[\hat{f}(x_0; D)]} + \text{Var}(\epsilon),$$

What is the expectation taken over?

- The expectation is over all sources of uncertainty.

- There are two sources of uncertainty:
  - The expected test error is for a random Y given a fixed x (Y is from the test set), and a random $\hat{f}$, which is estimate from the training data.
  - As such the expectation is over $Y|X = x$, and on the training set.



T/F?

$$\text{Bias}(\hat{f}(X)) = \mathbb{E}[\hat{f}(X)] - f(X)$$

The expectation is over the test set

$$\text{Bias}(\hat{f}(X)) = \mathbb{E}[\hat{f}(X)] - f(X)$$

The expectation is over the **training set**

$$\mathbb{E}_{D,\epsilon}\left[y_0 - \hat{f}(x_0; D)\right]^2 = \overline{\text{Bias}_D^2[\hat{f}(x_0; D)]} + \overline{\text{Var}_D[\hat{f}(x_0; D)]} + \text{Var}(\epsilon),$$

3. **Risk minimisation**

(a) Show that $\mathbb{E}[(Y - f(X))^2]$ is minimised by $f(x) = \mathbb{E}[Y \mid X = x]$.



(a) Show that $\mathbb{E}[(Y - f(X))^2]$ is minimised by $f(x) = \mathbb{E}[Y \mid X = x]$.

Interpret

Interpret what this expectation tells us

(a) Show that $\mathbb{E}[(Y - f(X))^2]$ is minimised by $f(x) = \mathbb{E}[Y \mid X = x]$.

- Risk function
- It's the expected prediction error when using a f(X) to predict Y.
- Here f is a known function (no estimation of f required)
- Called the MSE of the model f

- A function of X that gives the expected value of Y for each X=x
- It captures the systematic relationship between Y and X i.e.

$$Y = E(Y|X = x) + \epsilon$$

Problems for next week

For you to solve

1. **Bias and variance for simple linear regression** Assume that $Y = \theta^* + \epsilon$ where θ^* is an unknown parameter in $\mathbb{R}$, and ϵ is a standard normal random variable. Suppose that we are given $n \geq 1$ i.i.d. samples $y_1, \ldots, y_n$ according to distribution of Y.

   (a) Show that the sample mean $\bar{\theta} = (1/n) \sum_{i=1}^{n} y_i$ is obtained from

$$\bar{\theta} = \arg\min_{\theta \in \mathbb{R}} \sum_{i=1}^{n} (y_i - \theta)^2$$

   and that $\bar{\theta}$ is an unbiased estimator of θ^*.

   (b) Consider adding an ℓ_2 penalty term to define the parameter estimator as

$$\hat{\theta} = \arg\min_{\theta \in \mathbb{R}} \sum_{i=1}^{n} (y_i - \theta)^2 + \lambda \theta^2$$

   where λ is a non-negative parameter. Show that the estimator $\hat{\theta}$ is biased but has lower variance than $\bar{\theta}$. Provide the bias and variance of the estimator $\hat{\theta}$.

Is there any benefit of adding the second term in part (b)?

2. **Bias-variance decomposition** Assume that $Y = f(X) + \epsilon$, where ϵ is a random variable with zero mean and variance σ^2, independent of X. Let ℓ be the squared error loss function, i.e., $\ell(y, t) = (y - t)^2$. Show that the expected test loss, conditional on $X = x$ for some arbitrary input x, can decomposed as follows:

$$\mathbb{E}[\ell(Y, \hat{f}(X)) \mid X = x] = \text{Bias}(\hat{f}(x))^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

where $\hat{f}$ is estimated from the training data and $\text{Bias}(\hat{f}(x)) = \mathbb{E}[\hat{f}(x)] - f(x)$.

3. **Risk minimisation**

(a) Show that $\mathbb{E}[(Y - f(X))^2]$ is minimised by $f(x) = \mathbb{E}[Y \mid X = x]$.